Circumference of a circle

1

a 50.27 cm b 25.13 cm c 31.42 cm d 43.98 cm

2 169.6 cm

3 119.4 cm

4 2.23 cm

Arc length

5

a 12.6 units b 1.8 units c 6.2 m d 35.4 units

e 62.8 cm

6 $r = 19.76 \text{ cm}$

7 1.29 m

8 $25\,408.5 \text{ km}$

Area of circle

9

a 201.1 cm^2 b 78.5 cm^2 c 153.9 cm^2 d 113.1 cm^2

10

a $r = 10.59 \text{ cm}$ b 66.5 cm

Sectors

11

a 18.9 cm b 18.8 cm

12

a
- **i** 473.89 mm
- **ii** $11\,744.12\text{ mm}^2$

b
- **i** 218.75 cm
- **ii** 2683.95 cm^2

c
- **i** 92.40 m
- **ii** 497.59 m^2

d
- **i** 108.91 mm
- **ii** 730.62 mm^2

13

a $P = 257.73\text{ m}$

b $A = 3112.56\text{ m}^2$

14

a $P = 586.78\text{ cm}$

b $A = 15\,925.02\text{ cm}^2$

Segment

15 $6\pi - 9\sqrt{3}$ square units

16 $16\sqrt{3} + \dfrac{160\pi}{3}$ square units

17

a $\theta = 30^\circ$

b $108\pi - 324\text{ cm}^2$

c $1188\pi + 324\text{ cm}^2$

d $11\pi + 3 : \pi - 3$

18

a 73.7°

b 6.4 cm

c 4.1 cm^2

19 50.7 cm^2

20 $\dfrac{13357\pi}{360} - \dfrac{361}{2}\sin 37^\circ\text{ mm}^2$

21

a $\theta = \dfrac{240^\circ}{\pi}$

b 14.6 cm^2

Applications

22

a $\theta = 60^\circ$

b $\underline{10\pi + 3\sqrt{3}}$

23

a $x = 6$

b $\theta = 62^\circ$

c 9.8 cm^2

24

a $r = 12 \text{ cm}$

b $4\pi \text{ cm}$

25 22.1 laps

26

a 242.9 cm^2

b 242.9 cm^2

c 113.1 cm^2

d 146.1 cm^2

e 261.8 cm^2

f 40.8 cm^2

g 83.9 cm^2

h 176.5 units^2

i 19.6 cm^2

j 226.2 cm^2

27

a 60.68 cm

b 43.98 cm

c 42.85 m

d 35.00 cm

e 80.40 cm

f 31.42 cm

28

a 157.8 cm^2

b 9.4 cm^2

29

a 37.4 cm

b 35.7 mm

30 7.2 m

31

a 71 mm^2

b $\$284$

32 14.1 cm^2

33

a $r = 40.2 \text{ cm}^2$

b $r = 4.9 \text{ cm}$